Sampling and Representation

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

Teach survey arithmetic together with its scope.

Supply all collection plans, keep the four examples separate and retain the graph extension with independent task keys.

Keep one habit: Ask who could be selected, who answered and which claim the denominator supports.

MAIN LESSON · SECTION 1

Prepare the main lesson and its retained extension

> Use fictional records and schedule separate extension time.

Goal and preparation

Primary target: Define a question and target **population**

Teach Week 19 as a two-session investigation using fictional data. A fictional school wants to know which activity Grade 7 students prefer.

Surveying only the coding club is easy, but the question concerns every Grade 7 student. First define the **population**, selection **process**, question and **response** options. No real student data is required for this exercise.

Assess the three stated outcomes through the independent task, not by copying the worked result.

By the end, I can define a **population**, selection frame and non**response** report for a survey.

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Main plan

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Retrieval and readiness

Without opening the old lesson, explain one key idea from Missing Data. Give an example and a mistake that the idea helps you avoid.

Look for: Detect absent and unusable values

Reference example, not the only acceptable wording: 0 is a recorded allowed duration. The other two are unknown under this schema.

Without opening the old lesson, explain one key idea from Over**fitting** Intuition. Give an example and a mistake that the idea helps you avoid.

Look for: Compare **training** and validation **evidence** for model capacity

Reference example, not the only acceptable wording: Teacher explanation: Choose the line with validation MAE 3. The curve **training** MAE 0 does not compensate for validation MAE 8.

Without opening the old lesson, explain one key idea from Types of Classification Error. Give an example and a mistake that the idea helps you avoid.

Look for: Build a labelled confusion matrix from matched reference and **prediction** pairs

Reference example, not the only acceptable wording: Reference clear/predicted clear contains V1,V2: 2. Clear/blocked contains V3: 1. Blocked/clear contains V4,V5: 2. Blocked/blocked contains V6,V7,V8: 3. Matrix [[2,1],[2,3]].

6/10 = 60 percent among those **response**s.

▼ Materials and extension planning

Python 3 editor with a fresh local practice file

Paper trace table and the student workbook

Fictional **input** records from this lesson; no real student identifiers

Apply the connected idea: Graphs and honest scales

Schedule the retained Graphs and honest scales strand as a separate extension within this week. Use the fictional counts, complete text-chart code and five extension tasks; it is not the earlier Week18 Missing Data lesson.

Readiness answer

2.

Python 3 editor with a fresh local practice file

Paper trace table and the student workbook

Fictional **input** records from this lesson; no real student identifiers

▼ **Earlier wording, alternatives and picture transcripts**

Teach the learning cycle

Retrieval answer guidance

Missing Data

Overfitting Intuition

Types of Classification Error

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

A survey about school reading preferences includes only volunteers from a mystery club. Explain why 100 **response**s may still fail to represent the school. Propose a better sampling plan using fictional data.

Concept to explain

Week 19: Sampling and **Representation**

Term	Meaning
sample	A selected group of observations, or an outcome drawn from a distribution; the context identifies which meaning applies.
population	The full group of cases or people a study aims to describe.
representation	A way to encode information so a person or system can work with it.
sampling bias	A systematic mismatch between selected examples and the population the study intends to describe.

Prepare the class

This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Readiness answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach population, selection and response as separate stages

> Retain the graph strand as a labelled extension with its own records.

Introduce the population question

A fictional school wants to know which activity Grade 7 students prefer. Surveying only the coding club is easy, but the question concerns every Grade 7 student.

First define the **population**, selection **process**, question and **response** options. No real student data is required for this exercise.

Run the complete club-count script

```
rows = [{"group": "club", "choice": "Code"},
        {"group": "club", "choice": "Read"},
        {"group": "club", "choice": "Code"}]
code_votes = 0
for row in rows:
    if row["choice"] == "Code":
        code_votes = code_votes + 1
print(code_votes, len(rows))
```

Limit the resulting2/3 claim

The result 2 of 3 describes these fictional club **response**s. It cannot by itself estimate the whole grade reliably. A better plan starts from a complete grade roster, uses a fair selection method and records non**response**.

Random selection gives each listed student a chance; it does not guarantee every small **sample** is representative.

Collect the extra-club-response explanation

> **If the school adds 20 more club responses, which coverage problem remains?**

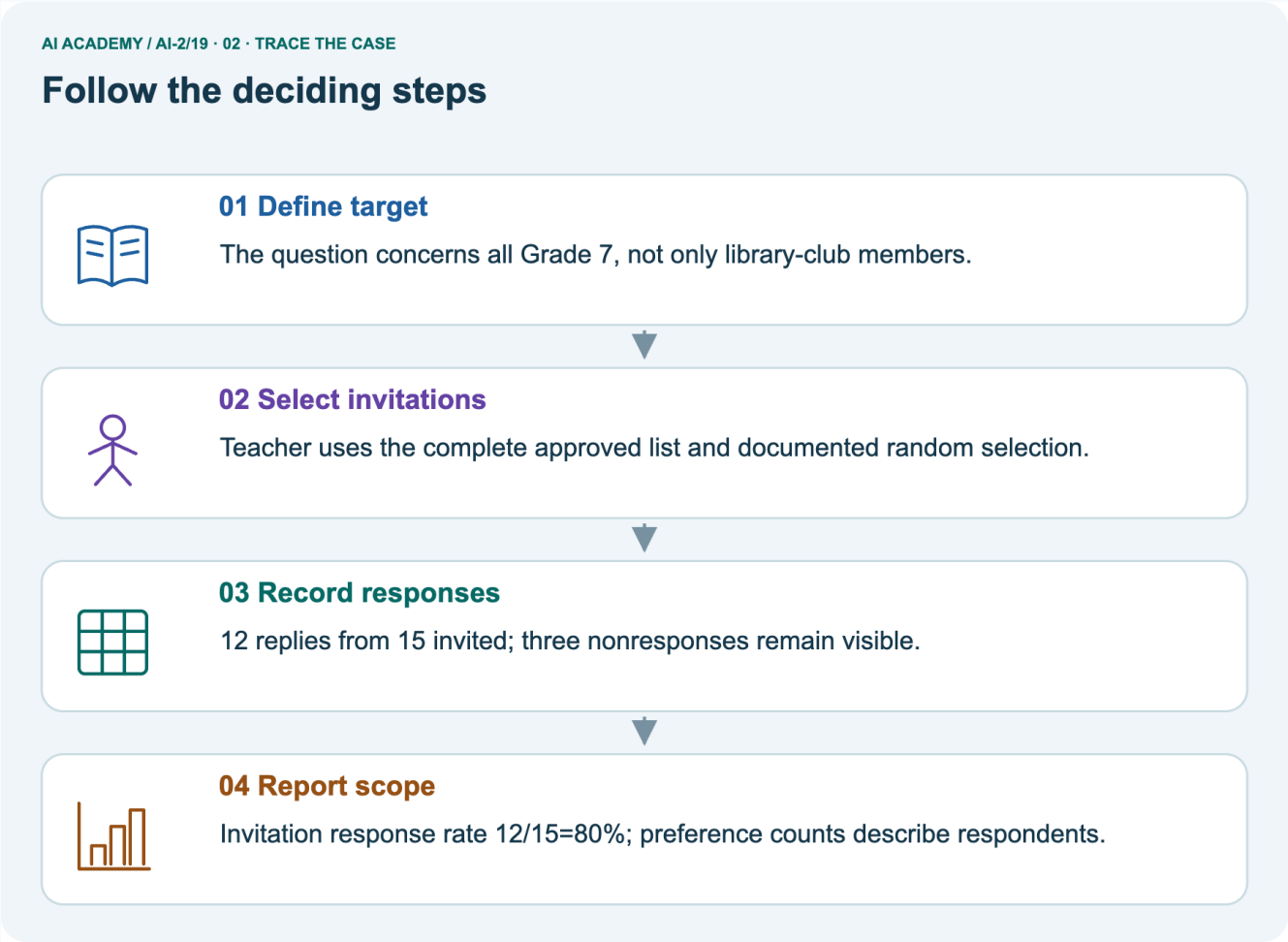
State the separate60/15/12 plan

A fictional Grade 7 class wants to understand book-format preferences across its whole grade.

Population: all 60 Grade 7 learners. Frame: current school-approved list of those 60, used only by the teacher.

Teacher randomly invites 15; 12 answer the same neutral comic/prose/either question. No names enter the analysis table.

Trace invitations, replies and nonresponse



Follow the labelled picture with the matching explanation. Open the vector diagram

A **sample** plan states who could be selected, how selection happens and what happens after invitation. Random invitations do not guarantee that respondents resemble those who declined.

Use a neutral question and preserve non**response** coverage. Learners can discuss aggregate counts without receiving the identifiable selection frame.

Even the random-invitation plan does not remove non**response** bias or justify revealing identifiable school lists.

Interpret the restricted-frame change

Change: Invite only the 15 library-club members instead.

Prediction: The invitation **process** no longer covers the whole grade through this frame.

Reason: Club members may differ in reading interests; selection is now limited to one group.

Elicit the picture explanation

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a **representation**; it does not document execution of a real AI model.

▼ Optional graph extension · Model and complete script

A fictional poll records 8 votes for Drawing and 10 for Building. A **bar chart** should make lengths proportional to counts, starting at zero.

A chart starting at 7 makes the visible bars 1 and 3 units tall and can exaggerate the difference.

```
counts = {"Drawing": 8, "Building": 10}
for label in counts:
    print(label, "#" * counts[label], counts[label])
```

The text chart uses one # per vote and a zero **baseline**. The second category has 2 more votes and 1.25 times as many votes, not three times as many.

Include the poll question, date or collection occasion and number of **response**s when presenting a real graph.

Difference 2 votes; ratio 6/4 = 1.5. Both bars begin at zero. Exit: No. Preference counts do not measure learning effectiveness or establish a cause.

▼ Earlier wording, alternatives and picture transcripts

Explain the mechanism

Input → **Process** → **Output**

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

1. Goal: books for the whole school.

2. **Sample**: one advanced-reading club at lunchtime.

3. Missing: younger readers, other interests, absent students, different access needs.

4. Repair collection across grades, times, interests, and access needs; then recheck.

Recalculate, retrace, or restate the reasoning in your own **representation**:

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

Concrete teaching case

Only adventure-club members answer a school reading survey.

Explain the sampling problem.

Exact explanation to model

Membership selects for a relevant interest. Include relevant grades and interests with appropriate data practices.

Explain the added worked example

Key terms

Term	Meaning
Population	The full group a question concerns
Sample	The observed subset of that group
Sampling frame	The list or source from which a sample is selected
Response	A recorded answer from a selected participant

Apply the connected idea: Graphs and honest scales

Model the reasoning aloud

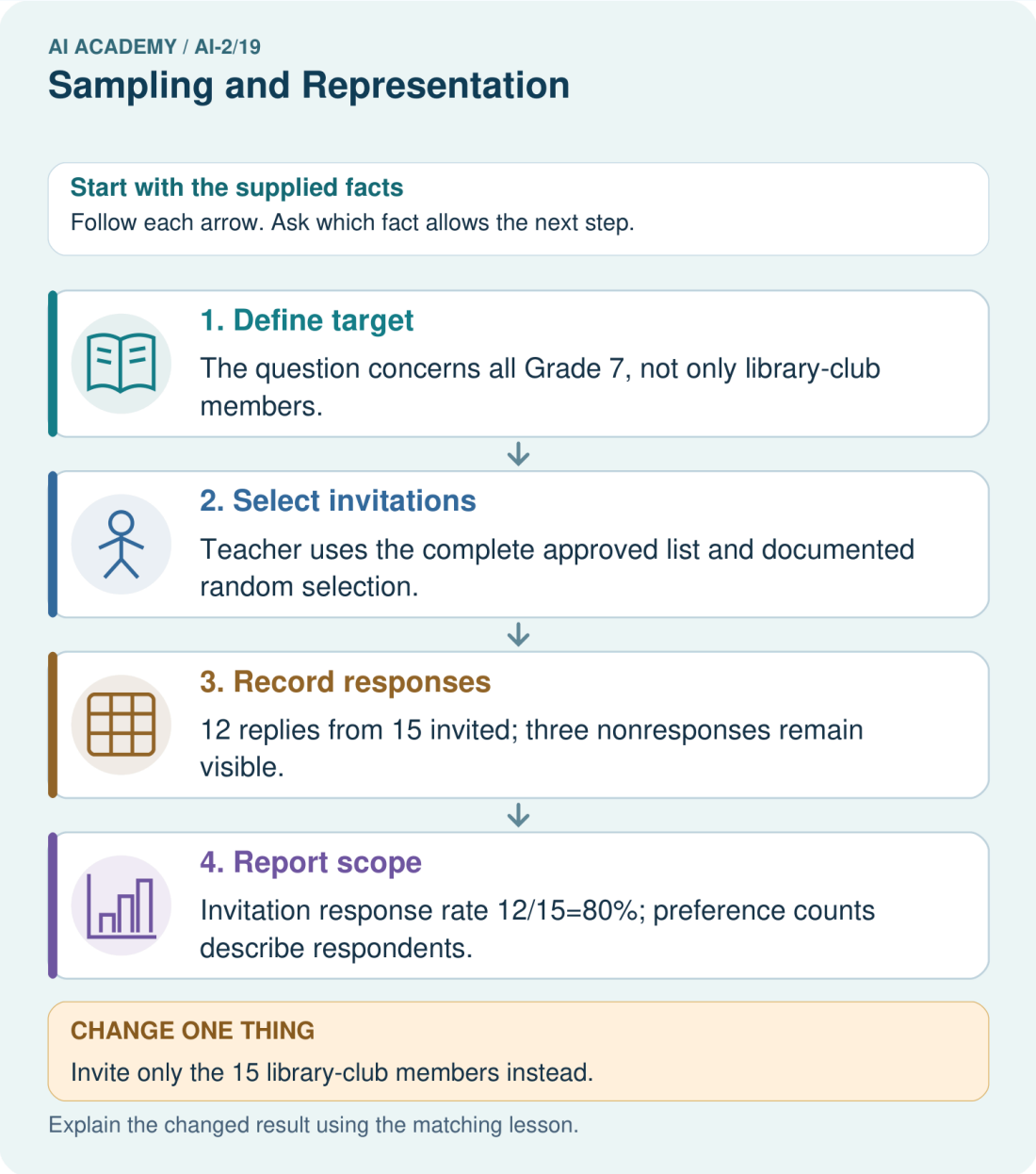
Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

A pictorial worked case: Sampling and Representation

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Define target	The question concerns all Grade 7, not only library-club members.
2. Select invitations	Teacher uses the complete approved list and documented random selection.
3. Record responses	12 replies from 15 invited; three nonresponses remain visible.
4. Report scope	Invitation response rate 12/15=80%; preference counts describe respondents.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Guide the main explanation, then fade support

> Collect each learner’s reasoning before revealing a key.

Main completion question and key

Completion task: Seven of 12 fictional respondents prefer longer lunch. State the result without generalizing.

Teacher check: 7/12 of these respondents prefer longer lunch, about 58.3 percent.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a **response** from every learner before discussion.

Main individual check

Use the worked case for Questions **population**s and **sample**s. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Selection and **response** are separate. Even a fair invitation **sample** can be distorted if only interested students respond. Report numbers invited and responding and explain who may be absent.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial question and changed case

Can **80% response** be reported as **80%** of the whole grade responding?

No. Twelve of fifteen invited is **80%**; twelve of sixty in the grade is 20%. Denominators answer different questions.

Changed-case reasoning: The invitation **process** no longer covers the whole grade through this frame. Club members may differ in reading interests; selection is now limited to one group.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Keep learners explaining

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

▼ Extension teaching cycle

Target: Choose an appropriate graph

Reserve further sessions for this topic; the 135-minute primary plan does not absorb all added content.

Completion practice: Counts are 12 and 15. State the difference and ratio using the true values.

Teacher check: Difference 3; ratio 15/12 = 1.25.

Individual check: For Graphs and honest **scale**s, use its worked case to evaluate this proposed explanation: "A successful run proves the conclusion". Explain the deciding step.

Expected correction: Counts alone do not show why students chose an activity or whether nonrespondents would choose differently.

Graph choices should match the question: categories suit bars; ordered time **observation**s can suit a line.

Apply the same support, independent transfer and delayed-review decisions to the named connected-topic workbook tasks.

▼ Earlier wording, alternatives and picture transcripts

Demonstrate and guide practice

Board sequence

Write the given facts: Only adventure-club members answer a school reading survey.

Write the question to explain: Explain the sampling problem.

Reveal after **prediction**s: Membership selects for a relevant interest. Include relevant grades and interests with appropriate data practices.

Use the earlier transfer question as optional practice. Reserve Plan independently for the first independent attempt before hints or feedback.

Classroom dialogue

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

Model these guided questions

Students outside the club are still not represented by the selection **process**. Exit: No. The 12 nonrespondents may differ. Report 8 of 20 and the limitation.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start to 7 minutes	Retrieve the prerequisite and explain the purpose in the opening section.
7 to 20 minutes	Read the worked example and diagram; ask learners to trace each step aloud.
20 to 33 minutes	Complete W1 and W2, then compare predictions with observed results.
33 to 45 minutes	Collect the guided prediction before running; discuss a mismatch and its cause.
Session 2 start to 8 minutes	Retrieve the main rule without showing the previous answer.
8 to 20 minutes	Use W3 to diagnose and repair the stated failure; rerun an earlier case.

Stage	Teacher action
20 to 35 minutes	Collect W4 independently, including code or plan, predictions and explanations, before hints.
35 to 45 minutes	Discuss W5 and the exit; use the fresh retry when support was needed.

Teach the connected idea

Stage	Teacher action
Session 1 start to 7 minutes	Retrieve the prerequisite and explain the purpose in the opening section.
7 to 20 minutes	Read the worked example and diagram; ask learners to trace each step aloud.
20 to 33 minutes	Complete W1 and W2, then compare predictions with observed results.
33 to 45 minutes	Collect the guided prediction before running; discuss a mismatch and its cause.
Session 2 start to 8 minutes	Retrieve the main rule without showing the previous answer.
8 to 20 minutes	Use W3 to diagnose and repair the stated failure; rerun an earlier case.
20 to 35 minutes	Collect W4 independently, including code or plan, predictions and explanations, before hints.
35 to 45 minutes	Discuss W5 and the exit; use the fresh retry when support was needed.

Reduce help deliberately

Individual reasoning check

Connected teaching cycle: Graphs and honest scales

Pictorial practice question

Reasoned teacher answer

Run the supplied main investigation

> Keep the selected independent task out of shared practice.

Supplied records and answer guidance

LabA: club-only coverage and3/15 nonresponse; response12/15=80%, whole-grade reply fraction12/60=20%. B: whole-frame random invitation with the same80% response and20% whole-grade reply fraction; nonresponse and small-sample limits remain. C: all60 invited but12/60=20% response,48 nonresponses; invitation coverage does not fix response bias. No activity-preference outcome is supplied. Main code2/3 describes club respondents only. The separate graph extension has8/10 votes, difference2 and ratio1.25. Reserve the new activity-survey plan for independent W4.

Predict, investigate and record

1. Predict → Test → Record → Explain
2. Keep expected and actual results separate. Surprises are useful data when recorded honestly.
3. Audit the supplied three fictional survey plans. State population, invitation frame, selection method, invited and responding counts, response denominator and the remaining coverage/nonresponse limits. Do not conduct a real survey for this exercise.
4. What did the hardest test teach you that the normal test did not?

Keep the evidence

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors

Record	What to capture
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Main misconceptions

A successful run proves the conclusion: Selection and **response** are separate. Even a fair invitation **sample** can be distorted if only interested students respond.

Report numbers invited and responding and explain who may be absent.

Matching the example is enough: Require the new W4 case, explicit **prediction**s and an explanation. Fresh retry: plan a **sample** about library opening times for the whole grade.

A library-club-only survey fails coverage; require a full-grade frame and non**response** plan.

A convenient shortcut preserves meaning: Changing the report heading from club **response**s to all students does not fix sampling. Change the collection plan or narrow the claim.

Avoid collecting names when anonymous counts are sufficient; a teacher can manage selection without publishing identities.

Choose support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.

What the response shows	Next teaching action
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ Extension misconceptions

A successful run proves the conclusion: Counts alone do not show why students chose an activity or whether nonrespondents would choose differently.

Graph choices should match the question: categories suit bars; ordered time **observation**s can suit a line.

Matching the example is enough: Require the new W4 case, explicit **prediction**s and an explanation. Fresh retry: Reading 4, Building 8 and Drawing 6; draw all bars from zero with one **scale**.

Building has twice Reading, and 2 more than Drawing.

A convenient shortcut preserves meaning: A graph with bars beginning at 7 turns counts 8 and 10 into visible lengths 1 and 3. Redraw from zero and label counts directly.

A line chart can use a restricted **axis** for detail if clearly labelled, but it must not disguise its **scale**.

▼ Earlier wording, alternatives and picture transcripts

Run the weekly evidence lab

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Make the lab runnable

Use the supplied main investigation for shared practice. Reserve Plan independently for the individual first attempt. The named extension has separate tasks and feedback.

Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

Check these misconceptions

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the selected main independent task

> Reveal its key after the first attempt; extension tasks have their own keys.

Selected independent question and key

Selected case: Design an anonymous **sample** plan to ask all Grade 7 about three activity choices. State **population**, **sampling frame**, selection method, question, and how you will report non**response**.

Use the full grade roster, teacher-managed random selection or a clearly described balanced plan, neutral identical options, anonymous **response** records and invited/responded counts.

Explain limits of non**response** and small **sample**s.

Judge reasoning and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one repair action

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Define the population

The question is whether all Grade 7 students prefer a longer lunch. Name the **population** and one unsuitable **sample**.

Population: all Grade 7 students. Only lunch-club members or one friend group is an unsuitable basis for a whole-grade claim.

W2 Read the sample result

Seven of 12 fictional respondents prefer longer lunch. State the result without generalizing.

7/12 of these respondents prefer longer lunch, about 58.3 percent.

The numerator is seven respondents choosing longer lunch and the denominator is all twelve observed respondents. The fraction is 7/12, approximately 58.3 percent.

Say “of these respondents”; the calculation alone supplies no **evidence** about the preferences of students who were not represented.

W3 Repair a claim

A report says all Grade 7 prefers coding after surveying only the coding club. Rewrite it.

Among surveyed coding-club respondents, the observed choices favored coding; the result does not establish the whole grade preference.

W5 Separate invitation and response

Twenty students are selected fairly but only 8 answer. Does fair selection remove the need to report non**response**?

No. The 12 nonrespondents may differ. Report 8 of 20 and the limitation.

▼ Extension questions and keys

Graphs and honest scales — W1 Read the scale

A bar is 6 squares long with 2 votes per square. How many votes is that?

12 votes.

One square represents two votes. Six equal squares therefore represent 6 times 2 = 12 votes. Ask the learner to name both the visible length and the unit **scale** before giving the count.

Graphs and honest scales — W2 Compare fairly

Counts are 12 and 15. State the difference and ratio using the true values.

Difference 3; ratio 15/12 = 1.25.

The count difference is 15 - 12 = 3 votes. Taking the larger count relative to the smaller gives 15/12 = 1.25, so 15 is 1.25 times 12, or 25 percent larger. Label which count is the reference for the ratio.

Graphs and honest scales — W3 Repair a misleading graph

Bars for 12 and 15 start at 10. Explain the misleading lengths and repair.

Visible lengths 2 and 5 suggest a 2.5 ratio. Start at zero, use a common **scale** and label counts 12 and 15.

Graphs and honest scales — W4 Draw independently

Draw a **bar chart** for Music 6, Art 9, Coding 3. Include title, category labels, vote units, zero **baseline** and equal **scale**. State two supported comparisons.

Music 6, Art 9, Coding 3 on one **scale**. Art has 3 more than Music and three times Coding. Accept other correct comparisons.

Graphs and honest scales — W5 Limit the claim

Art has the most votes among 18 **response**s. Does that establish Art is best for learning?

No. Preference counts do not measure learning effectiveness or establish a cause.

▼ Earlier wording, alternatives and picture transcripts

Assess and explain the answers

Worked case answer

Membership selects for a relevant interest. Include relevant grades and interests with appropriate data practices.

Additional transfer practice

You survey only the chess club about all students' favourite books. Identify the sampling limitation and one improvement.

Reasoned practice answer

Chess-club readers may not represent all students. Include learners from other groups and reading interests; broader coverage still needs honest reporting of who responded.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support

Criterion	0 points	1 point	2 points
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “A convenient **sample** represents everyone automatically.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Club membership and volunteering select for interests. Include relevant grade groups and varied reading interests using a planned sampling approach, record who was invited and who responded, and disclose missing groups.

More **response**s from the same selected group do not automatically fix coverage.

Explanations for the additional workbook examples

W1 Define the population

W2 Read the sample result

W3 Repair a claim

W4 Plan independently

Design an anonymous **sample** plan to ask all Grade 7 about three activity choices. State **population**, **sampling frame**, selection method, question, and how you will report non**response**.

Use the full grade roster, teacher-managed random selection or a clearly described balanced plan, neutral identical options, anonymous **response** records and invited/responded counts.

Explain limits of non**response** and small **sample**s.

W5 Separate invitation and response

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, support and return

> Keep main and extension assessment evidence distinguishable.

Project record

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Document whose requests inform the recommender and avoid treating an easily reached group as all intended users.

Main support and fresh assessment

Use fictional roster cards. Extension: draw several small random **sample**s from a known fictional **population** and compare their proportions.

Independent W4 **evidence**: Use the full grade roster, teacher-managed random selection or a clearly described balanced plan, neutral identical options, anonymous **response** records and invited/responded counts.

Explain limits of non**response** and small **sample**s. Record whether the learner needed no help, a hint or explanation. Fresh retry: plan a **sample** about library opening times for the whole grade.

A library-club-only survey fails coverage; require a full-grade frame and non**response** plan.

Delayed return

Suggested return points in this level: ai-2/20, ai-2/22, ai-2/25. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent **response** and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Bridge to next week

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Next: Week 20: Repair Bias in Data. Retrieve today's **evidence** and identify what carries forward and what is new.

Next is Week20: Repair Bias in Data. Carry forward the intended population, invitation frame, response process and documented coverage limits when judging proposed repairs.

▼ Extension support and reassessment

Use graph paper with a marked zero. Extension: show the same values on two clearly labelled **scale**s and discuss visual emphasis.

Independent W4 **evidence**: Music 6, Art 9, Coding 3 on one **scale**. Art has 3 more than Music and three times Coding. Accept other correct comparisons. Record whether the learner needed no help, a hint or explanation.

Fresh retry: Reading 4, Building 8 and Drawing 6; draw all bars from zero with one **scale**. Building has twice Reading, and 2 more than Drawing.

▼ Earlier wording, alternatives and picture transcripts

Connect and extend learning

Transparent Book Recommender

Test reader profiles across reading levels, genres, ages, formats, and accessibility needs.

Support and stretch

Trace one record or condition at a time with counters and labelled columns. Stretch: introduce a tie, boundary or missing field and declare its handling policy before rerunning.

Exit reflection

Bridge to the next week

Support and fresh reassessment

Readiness and reassessment for Graphs and honest scales

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative teaching checklist**

REFERENCE · SECTION 7

Reference · Alternative teaching checklist

> Use as a reminder alongside the main route.

Planning reference

Teaching target: Define a question and target **population**

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual **response**s. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

Before class, solve the supported practice: Seven of 12 fictional respondents prefer longer lunch. State the result without generalizing.

Reasoned practice answer: 7/12 of these respondents prefer longer lunch, about 58.3 percent.

Check every learner before discussion: Use the worked case for Questions **population**s and **sample**s. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Selection and **response** are separate. Even a fair invitation **sample** can be distorted if only interested students respond. Report numbers invited and responding and explain who may be absent.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Invite only the 15 library-club members instead. Require a **prediction** and a reason before comparing. Expected result: The invitation **process** no longer covers the whole grade through this frame.

Reason: Club members may differ in reading interests; selection is now limited to one group.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback and retained evidence**

REFERENCE · SECTION 8

Reference · Feedback and retained evidence

> Keep supported practice separate from independent evidence.

Record and repair

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.
</details>

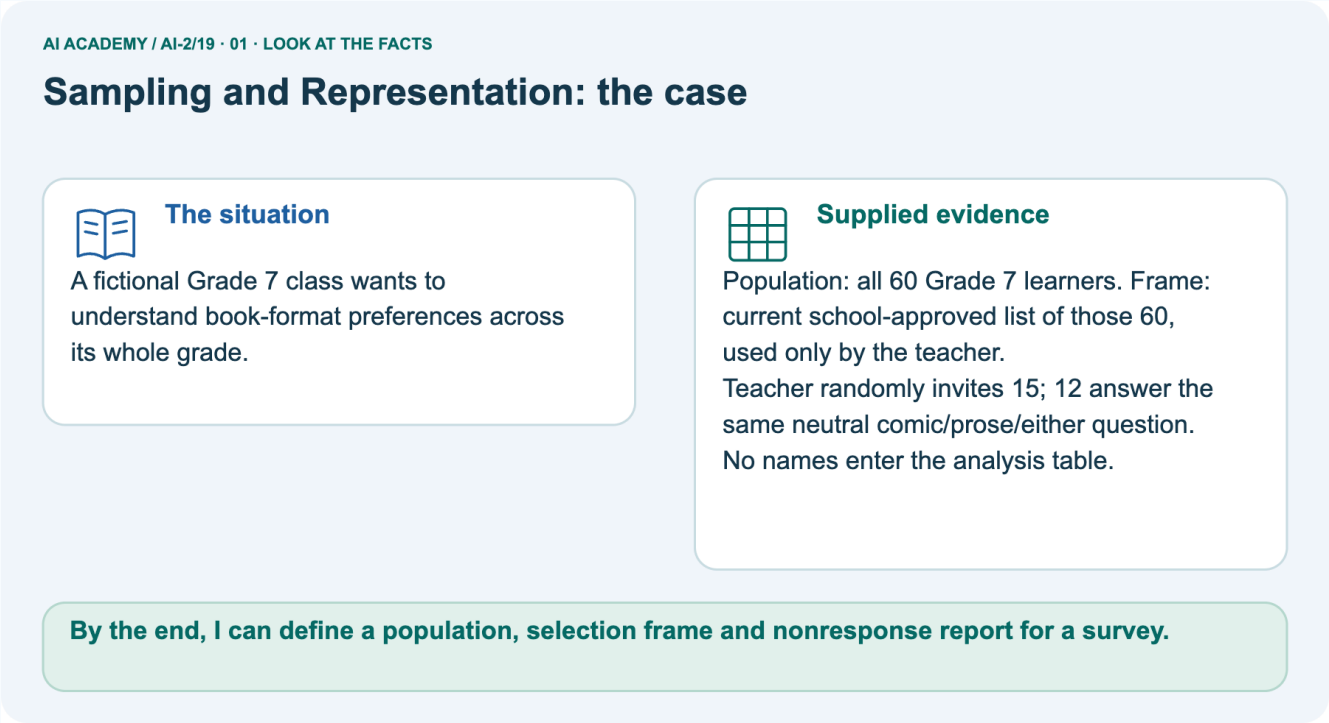
<details>
<summary>Reference · Teach with the six pictures</summary>
</details>

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose a picture for a specific explanation.

01. Sampling and Representation: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

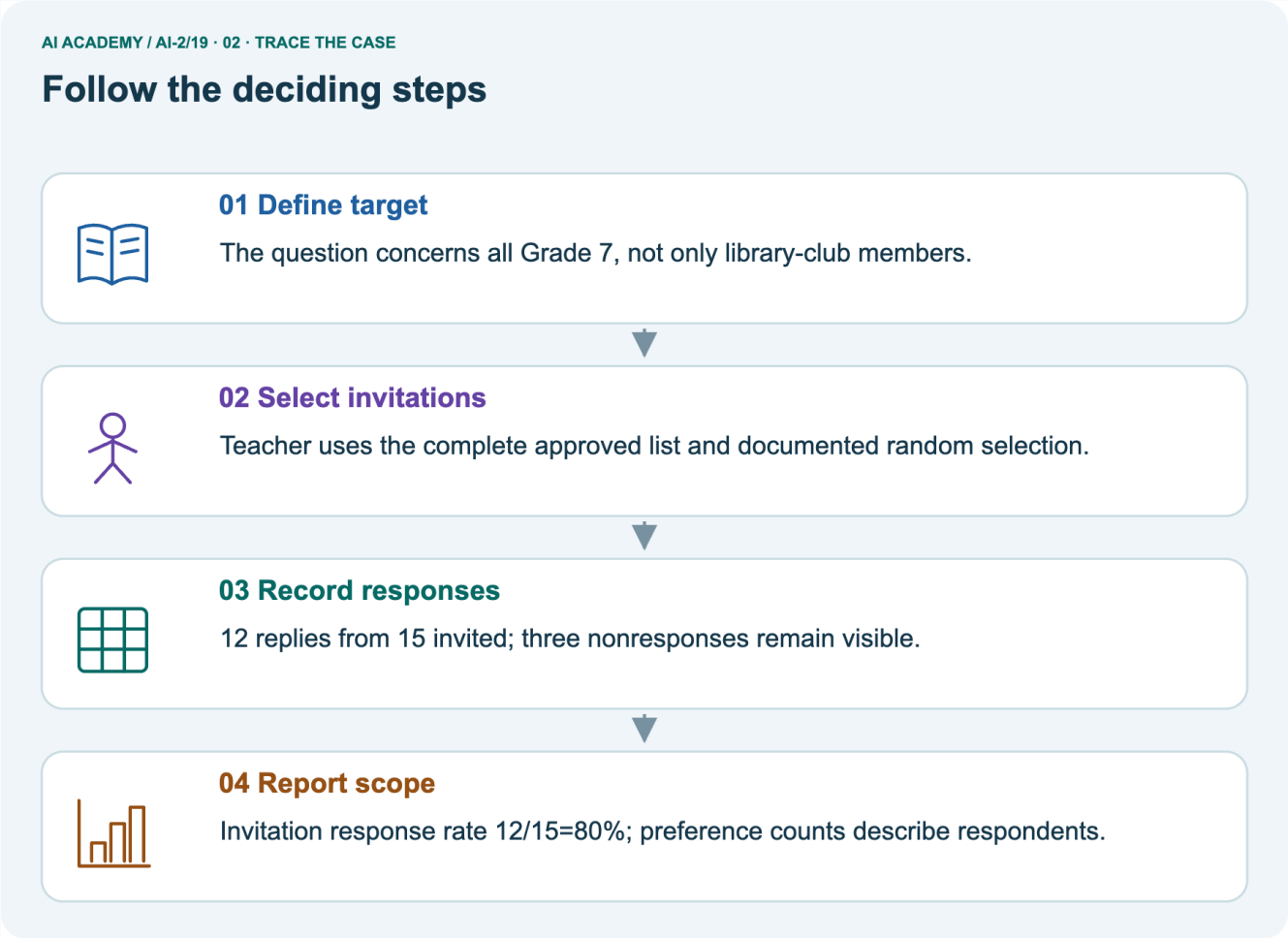
A fictional Grade 7 class wants to understand book-format preferences across its whole grade. Supplied facts: **Population**: all 60 Grade 7 learners. Frame: current school-approved list of those 60, used only by the teacher.

Teacher randomly invites 15; 12 answer the same neutral comic/prose/either question. No names enter the analysis table.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



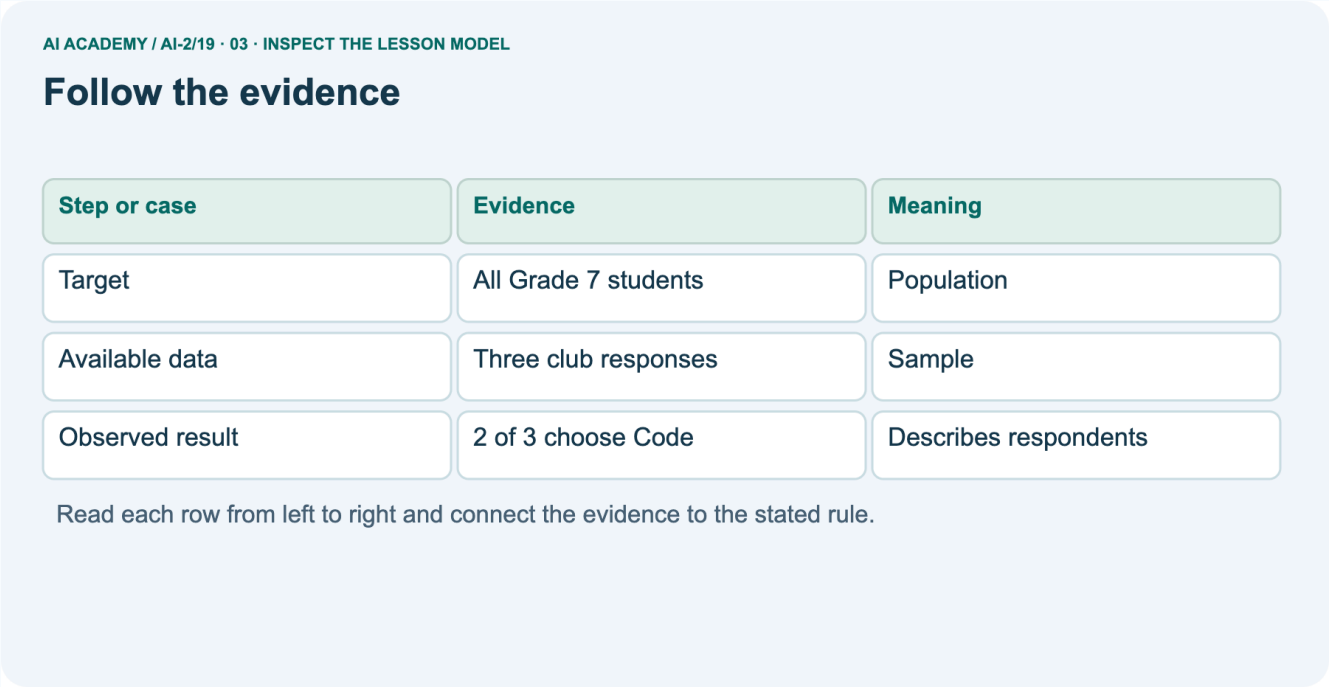
Follow the labelled picture with the matching explanation. Open the vector diagram

Define target: The question concerns all Grade 7, not only library-club members. → Select invitations: Teacher uses the complete approved list and documented random selection. → Record **response**s: 12 replies from 15 invited; three non**response**s remain visible. → Report scope: Invitation **response** rate 12/15=**80%**; preference counts describe respondents.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Follow the evidence



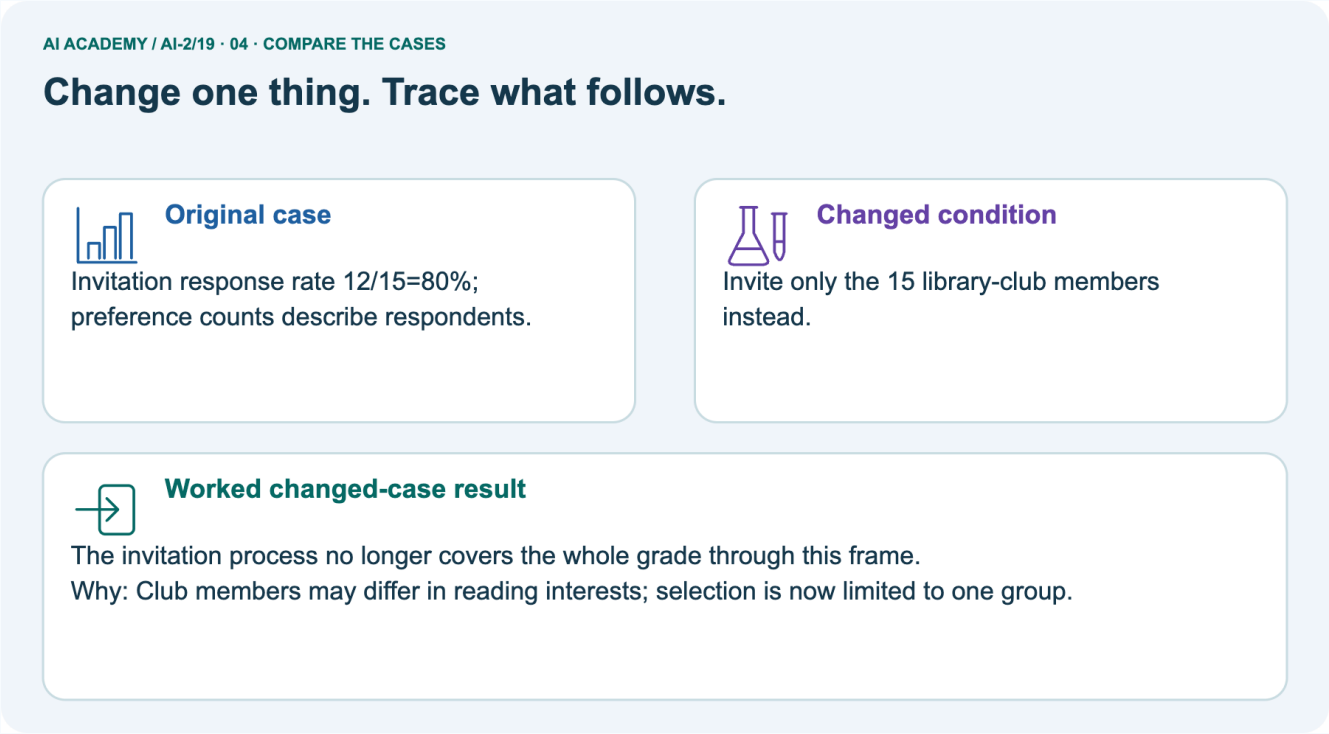
Follow the labelled picture with the matching explanation. Open the vector diagram

Follow the **evidence**. Target | All Grade 7 students | **Population**. Available data | Three club **response**s | **Sample**. Observed result | 2 of 3 choose Code | Describes respondents

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Invitation **response** rate 12/15=**80%**; preference counts describe respondents.
Changed condition: Invite only the 15 library-club members instead.

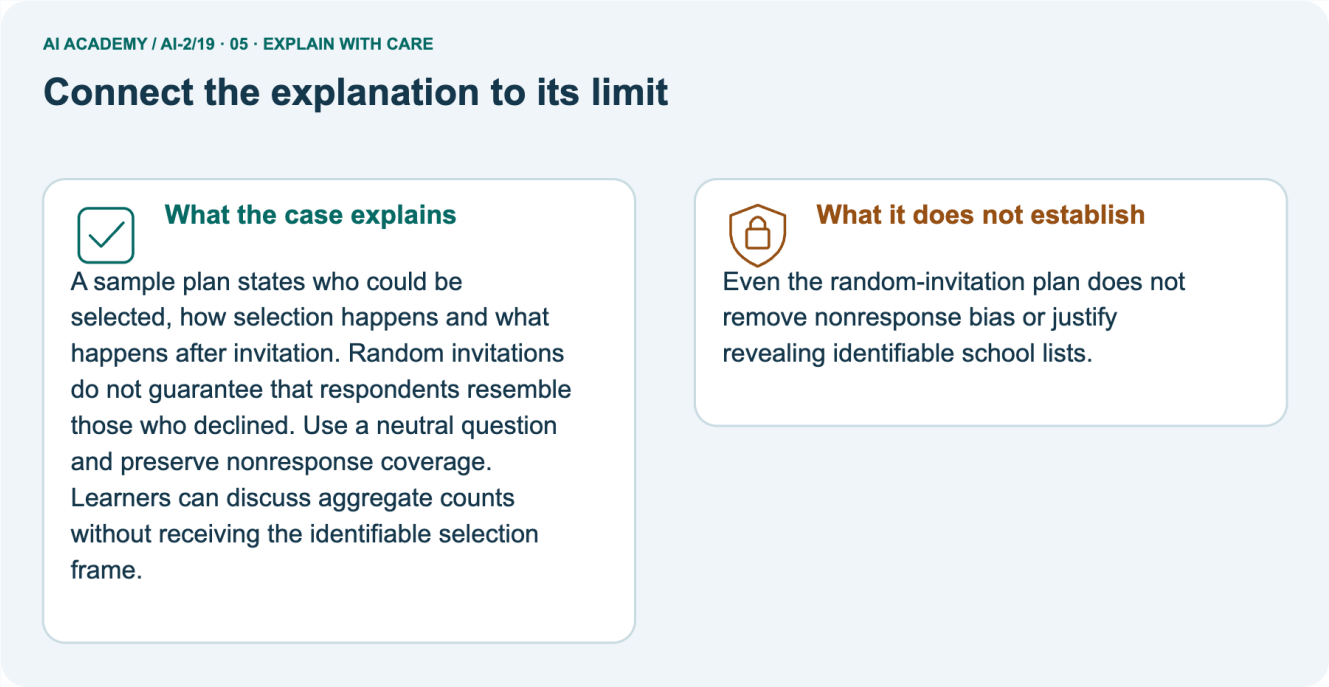
Expected result: The invitation **process** no longer covers the whole grade through this frame.
Reason: Club members may differ in reading interests; selection is now limited to one group.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Club members may differ in reading interests; selection is now limited to one group.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A **sample** plan states who could be selected, how selection happens and what happens after invitation. Random invitations do not guarantee that respondents resemble those who declined.

Use a neutral question and preserve non**response** coverage. Learners can discuss aggregate counts without receiving the identifiable selection frame.

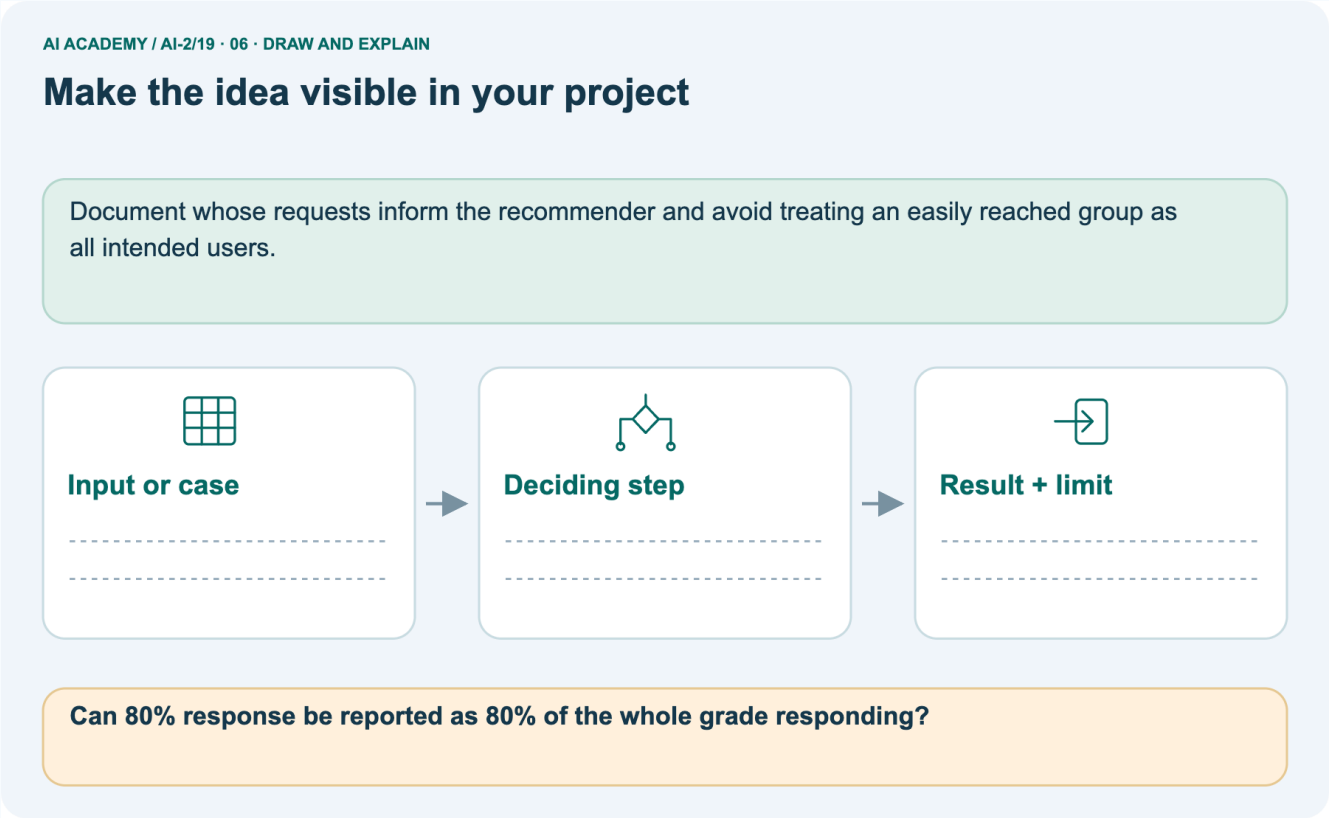
Limit: Even the random-invitation plan does not remove non**response** bias or justify revealing identifiable school lists.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Even the random-invitation plan does not remove non**response** bias or justify revealing identifiable school lists.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Document whose requests inform the recommender and avoid treating an easily reached group as all intended users. Draw your **input**, deciding step, result and limit.

Supported question: Can **80% response** be reported as **80%** of the whole grade responding?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: No. Twelve of fifteen invited is **80%**; twelve of sixty in the grade is 20%. Denominators answer different questions.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Sampling and Representation: the case

02. Follow the deciding steps

03. Follow the evidence

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.